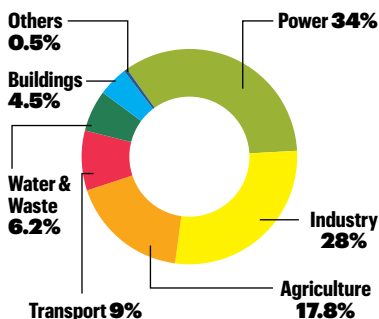




PHOTO BY GETTY IMAGES

CARBON TRACKER

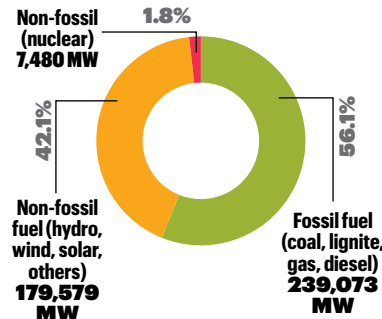
- The power sector emits the most, especially coal-fired power plants
- Among industries, cement, steel, and textiles are the biggest emitters



SOURCE DECARBONISING INDIA: CHARTING A PATHWAY FOR SUSTAINABLE GROWTH; 2019

FOSSIL RECORD

- Fossil fuels include coal, lignite, gas, and diesel
- Non-fossil fuels include hydro, wind, solar, and others



SEGMENT-WISE CARBON EMISSION CONTRIBUTION TO INDIA'S POWER SECTOR; DATA AS OF NOVEMBER 30, 2023; **SOURCE** MINISTRY OF POWER

INDIA'S NET-ZERO COMMITMENT

- Net zero by 2070
- Reduce emissions intensity of GDP by 45% by 2030, from the 2005 level
- Achieve 50% cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030

- In the Union Budget 2023-24, nearly \$4.3 billion (₹35,740 crore) was allocated for priority capital investments towards energy transition initiatives, net zero objectives, and energy security targets

SOURCE BT RESEARCH

Currently, green hydrogen costs more than energy from burning fossil fuels. But India's efforts to develop green hydrogen production capabilities, bolstered by the 2021 National Green Hydrogen Mission that aims to reduce fossil fuel imports by ₹1 lakh crore and cut CO₂ emissions by 50 million tonnes a year by 2030, is expected to help industrial decarbonisation significantly.

Aluminium is one of the metal industries that needs large amounts of power 24x7 to run efficiently and needs incentives for the switch, says John Slaven, CEO of Vedanta Aluminium. "There is a need to incentivise the usage of renewable energy through subsidies and scale up policies supporting its adoption. A focus on energy efficiency programmes and the expansion of low-carbon technologies like green hydrogen will play a crucial role," says Slaven.

Industries that need power 24x7 require efficient power storage technologies. This will enhance the suitability of renewable energy sources such as wind and solar and make them more effective for integration into industrial operations.

Florence Verzelen, Executive VP of software solutions provider Dassault Systemes, says life-cycle analysis using a digital twin of a manufacturing process could help develop the required technology. "Supporting the implementation of life cycle analysis in product and process development... can be helpful," says Verzelen.

SUSTAINABLY BUILDING INDIA

Urban India is growing at a fast clip, which means more housing, more roads, and more flyovers. Urban planners must keep in mind key sources of greenhouse gases, such as fossil fuel use in transportation and unsustainable economic practices while looking into the future.